

Systematic Review of IgA Nephropathy Coexisting With Alport Syndrome



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Introduction: In recent years, IgA nephropathy (IgAN) coexisting with Alport syndrome (AS) has been increasingly reported. However, the clinicopathological and molecular features, treatment, and prognosis of this entity have not been fully elucidated.

Methods: We first retrospectively analyzed pediatric patients with IgAN coexisting with AS (IgAN-AS) admitted to our center, and performed galactose-deficient IgA1 (Gd-IgA1) immunostaining on renal biopsy specimens. We then systematically searched PubMed, Scopus, Web of Science, Wanfang Data, and the China National Knowledge Infrastructure for publications reporting on patients with IgAN-AS.

Results: A total of 81 patients from 72 families were included. Microscopic hematuria was a universal finding. Nephrotic-range proteinuria was significantly more frequent in children than in adults (73.7% vs. 8.8%). Pathological findings confirmed the dual diagnosis, with mesangial IgA-dominant deposits and significant glomerular basement membrane abnormalities, including thinning (69.5%, 41/59) and typical AS-related changes (28.8%, 17/59). Genetic analysis identified 66 distinct pathogenic variants across *COL4A3/A4/A5* genes in 77 patients. Immunologic findings were heterogeneous, with inconsistent Gd-IgA1 deposition or circulating levels. Among patients evaluated for treatment response, complete remission (CR) and partial remission (PR) were observed in 7.1% (1/14) and 64.3% (9/14), respectively, during immunosuppressive treatment. In 27 patients with follow-up data, the median age at last visit was 16.0 (10.0–24.0) years, and 11 (40.7%) reached the kidney end point.

Conclusion: IgAN-AS appears to be a heterogeneous condition with potentially unfavorable kidney outcomes. Some patients exhibited both immune and structural injury, suggesting that biomarkers with high sensitivity and specificity for IgAN may aid in disease stratification.

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KEYWORDS: Alport syndrome; coexistence; galactose-deficient IgA1; glomerular basement membrane; IgA nephropathy; systematic review

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IgAN is the most prevalent primary glomerulonephritis worldwide and remains a leading cause of chronic kidney disease (CKD) and kidney failure.¹ The clinical spectrum of IgAN is highly variable, ranging from asymptomatic microscopic hematuria to rapidly progressive glomerulonephritis.² IgAN is an immune complex-mediated disease in which Gd-IgA1 plays a central role in its pathogenesis.² AS is one of the most common inherited kidney diseases, caused by mutations in the *COL4A3*, *COL4A4*, or *COL4A5* genes.³ A pathognomonic finding of AS on kidney biopsy is a

lamellated glomerular basement membrane (GBM), and its clinical spectrum ranges from isolated hematuria to kidney failure, sometimes with accompanying sensorineural deafness and ocular anomalies.³

Recent studies have shown that the coexistence of AS in patients with either familial or sporadic IgAN is not uncommon.^{4–6} However, the current understanding of IgAN-AS remains limited. It is unclear whether the coexistence of IgAN and *COL4A3/A4/A5* variants is coincidental or whether abnormalities in the GBM predispose to mesangial IgA deposition. In addition, it remains uncertain whether IgAN-AS should be categorized as primary IgAN or as a secondary consequence of mechanical entrapment due to defective GBM. Although immunosuppressive therapy is often required for progressive IgAN, patients with AS are primarily managed with renin-angiotensin system blockade (RASB) as the standard

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of care. The treatment response and optimal therapy for IgAN-AS are unclear. Finally, though both IgAN and AS are chronic progressive diseases, the prognosis and associated risk factors for patients with IgAN-AS have not yet been clearly elucidated.

Therefore, to address these uncertainties and ultimately improve clinical management, we first analyzed a case series of patients with IgAN-AS from our own center and subsequently conducted a systematic review to better characterize the clinicopathological and molecular features, treatment, and prognosis of IgAN-AS patients.

METHODS

Institutional Case Series

At our institution, immunofluorescence staining for the $\alpha 3$ and $\alpha 5$ chains of type IV collagen and electron microscopy have been routinely performed on all pediatric renal biopsy specimens since 2015. We retrospectively reviewed our electronic medical records to identify all children (aged < 18 years) diagnosed with IgAN-AS between January 1, 2015, and December 31, 2023. We collected and analyzed data on the clinical, pathological, genetic, therapeutic, and prognostic features of these patients. The final date of follow-up was June 1, 2025.

Glomerular Gd-IgA1 and IgA deposition were assessed in paraffin-embedded kidney sections from children with IgAN-AS, classic IgAN (positive control), and AS (negative control) at our center using double immunofluorescence staining, generally following the protocol described by Suzuki *et al.*⁷ The sections were examined with a fluorescence microscope (Nikon, Japan). Immunofluorescence intensity was semiquantitatively scored, and representative images were captured.

Systematic Review

Search Strategy, Selection Criteria, and Data Extraction

We searched 5 electronic databases, namely PubMed, Scopus, Web of Science, Wanfang Data, and the China National Knowledge Infrastructure. The search strategy was as follows: ("IgA nephropathy" OR "IgA deposits" OR "IgA deposition") AND ("Alport syndrome" OR *COL4A3* OR *COL4A4* OR *COL4A5*). In addition, the reference lists of relevant articles and reviews were manually screened to identify further eligible studies. The search was restricted to publications in English and Chinese, and the final search was conducted up to August 2025. Two investigators independently performed the literature search, screening, and selection. Discrepancies were resolved by a third researcher or consensus-based discussion.

Eligible studies were those reporting patients who met the diagnostic criteria for IgAN-AS. IgAN was defined as dominant or codominant mesangial IgA deposition on renal immunofluorescence in the first biopsy. In the absence of detailed immunofluorescence data, cases were included only if the diagnostic criteria for IgAN were explicitly stated by the original authors and were consistent with the criteria described above. AS was diagnosed in patients presenting with glomerular hematuria and/or proteinuria if ≥ 1 of the following criteria was fulfilled^{6,8,9}: (i) GBM abnormalities on electron microscopy, including irregular thinning and thickening, and lamellation and splitting in lamina densa; (ii) abnormal staining of type IV collagen α chains in glomerular or epidermal basement membranes; or (iii) identification of pathogenic (P) or likely pathogenic (LP) variants in *COL4A3*, *COL4A4*, or *COL4A5*, as defined by the American College of Medical Genetics and Genomics (ACMG) and the Association for Molecular Pathology guidelines.¹⁰ The exclusion criteria were as follows: (i) the presence of additional P/LP variants in other genes, (ii) patients lacking both detailed immunofluorescence data and clearly specified diagnostic criteria for IgAN, and (iii) diagnosis of AS based solely on *COL4A3/A4/A5* variants classified as variant of uncertain significance.

For variants lacking detailed ACMG class and supporting criteria in the original reports, automated annotation tools (VarSome¹¹ and Franklin [<https://franklin.genoox.com>]) were consulted to facilitate evidence collection. Final ACMG class and supporting criteria were determined according to the ACMG/Association for Molecular Pathology guidelines and remained consistent with the conclusions of the original studies. It should be noted that the *COL4A3* c.4793T>G (p.L1598R) was classified as LP in 1 study (supporting ACMG criteria: PS1+PP2+PP3)⁵ and as variant of uncertain significance in another study,¹² where the supporting ACMG criteria were not provided. Given the confirmed in-trans occurrence of *COL4A3* c.4793T>G (p.L1598R) with pathogenic variants in multiple independent genetically confirmed autosomal recessive AS (ARAS) cases (fulfilling PM3 at the strong level),¹³ together with its rarity in population databases (PM2) and supporting computational evidence (PP2 and PP3), this variant is classified as LP.

We extracted the following information for each patient: age at onset, age at diagnosis, sex, degree of hematuria, level of proteinuria, serum creatinine, estimated glomerular filtration rate, serum albumin, serum total cholesterol, renal biopsy findings, genetic testing results, molecular findings, treatment regimens with immunosuppressive agents and RASB, changes in proteinuria after treatment, duration of follow-up, and

renal outcomes. For patients without formally reported Oxford classification scores, the individual lesions—endocapillary hypercellularity (E), segmental glomerulosclerosis (S), tubular atrophy/interstitial fibrosis (T), and crescents (C)—were retrospectively classified according to the Oxford criteria when sufficient detail was available in the pathological descriptions.¹⁴ Mesangial hypercellularity (M) was not assessed in these cases because the quantitative threshold required for M0/M1 scoring could not be reliably determined from qualitative descriptions. Data from other affected family members were collected if available. In addition, online supplementary materials were carefully reviewed to avoid missing relevant information, and cross-checking was performed to exclude duplicate cases. In studies lacking posttreatment proteinuria data, treatment response was categorized according to the authors' original descriptions. For patients who underwent repeat kidney biopsies, only the findings from the first biopsy were used for the summary of renal pathological features.

Definitions

According to the recent guidelines,^{6,8} AS was classified into 4 subtypes based on genetic findings as follows: (i) X-linked AS (XLAS, P/LP variant in *COL4A5*), (ii) ARAS (compound heterozygous or homozygous P/LP variants in *COL4A3* or *COL4A4* with both parents as carriers), (iii) *COL4A3/COL4A4* heterozygotes (single P/LP variant in *COL4A3* or *COL4A4*), and (iv) digenic AS (P/LP variants involving 2 of *COL4A3*, *COL4A4*, or *COL4A5*). Accordingly, patients with IgAN-AS were classified into 4 subgroups as follows: (i) IgAN-XLAS; (ii) IgAN-ARAS; (iii) IgAN with a single P/LP *COL4A3* or *COL4A4* variant (IgAN-*COL4A3* or IgAN-*COL4A4*); and (iv) IgAN-digenic AS.

Nephrotic-range proteinuria was defined as > 3.5 g/d or urine protein-to-creatinine ratio ≥ 3.5 g/g (350 mg/mmol) in adults, and as urine protein-to-creatinine ratio ≥ 2 g/g (200 mg/mmol) in spot urine or ≥ 1000 mg/m²/d or ≥ 50 mg/kg/d in children,¹⁵ or as explicitly stated as nephrotic-range proteinuria in the publication. Nephrotic syndrome was defined as nephrotic-range proteinuria with hypoalbuminemia (serum albumin < 30 g/l) or edema if albumin data were unavailable. Steroid-resistant nephrotic syndrome was defined as failure to achieve CR after 4 weeks of daily prednisone or prednisolone at standard dose.

Treatment response was evaluated in patients with baseline proteinuria ≥ 1.0 g/d or a urine protein-to-creatinine ratio ≥ 1.0 g/g; in pediatric patients, proteinuria ≥ 50 mg/kg/d was considered equivalent for response evaluation, even if the absolute

proteinuria did not reach these thresholds. CR was defined as proteinuria < 0.15 g/d (or urine protein-to-creatinine ratio < 0.2 g/g). PR was defined as either a $\geq 50\%$ reduction in proteinuria from baseline or a qualitative description of PR of proteinuria found in the source article. A reduction of $< 50\%$ was considered no remission. CKD was classified based on the 2024 Guideline for CKD from the Kidney Disease: Improving Global Outcomes organization.¹⁶ The kidney end point was defined as impaired kidney function with an estimated glomerular filtration rate < 60 ml/min per 1.73 m² or end-stage kidney disease.

Statistical Analysis

Statistical analyses were performed using R software 4.4.2 (<https://www.R-project.org/>). Continuous variables were reported as median (Q1–Q3) or mean $\pm$ SD, as appropriate. Categorical data were expressed as frequency (percentage). Categorical variables were compared between independent groups using chi-square test or Fisher exact test, as appropriate. A *P*-value < 0.05 was considered statistically significant.

RESULTS

Institutional Case Series

Over a 9-year period, our center diagnosed 558 pediatric patients with primary IgAN, of whom 5 (0.90%) were identified with IgAN-AS. Detailed clinical, pathological, genetic, treatment, and prognostic data for these 5 patients with IgAN-AS are presented in [Tables 1](#) and [2](#). All 5 children received immunosuppressive therapy. Cases 2 and 3 presented with nephrotic-range proteinuria and both achieved PR, although case 3 experienced frequent relapses. All 5 children received long-term RASB therapy. At the end of follow-up, cases 1 and 2 remained in CKD G1, whereas cases 3, 4, and 5 progressed to CKD G3–G4.

Four of the 5 unrelated patients had a family history of kidney disease, whereas the remaining patient (case 2) carried a *de novo* mutation ([Table 1](#)). The paternal aunt of case 1 harbored the same mutation and presented with more severe proteinuria (> 1.0 g/d), a histopathological diagnosis of focal segmental glomerulosclerosis (not otherwise specified), and ultrastructural changes typical of AS. Despite long-term RASB therapy, she experienced an earlier onset of renal function decline than case 1. In contrast, the mother of case 3 carried the same mutation but exhibited a much milder phenotype (microscopic hematuria and mild proteinuria) and a significantly better prognosis; she was treated with RASB without renal biopsy. Case 4's younger brother, who carried the same mutation, presented with nephrotic syndrome at a younger age but was not biopsied. Finally, the unbiopsied twin

Table 1. Clinical features of 5 children with IgAN-AS at our center

Patient	Sex	Family history	Age at onset, yrs	Clinical information at admission	Immunosuppressive drugs treatment	Outcome
1	Female	Father died of uremia in his 20s; paternal aunt with microscopic hematuria and proteinuria developed ESKD at 37 years.	15	Urine RBC: 122/ μ l; urine protein: 0.61 g/d; serum albumin and creatinine were normal.	Prednisone for 6 mos. Response: urine protein reduced to < 0.3 g/d.	Normal eGFR at 24 yrs
2	Female	No	10	Urine RBC: 1786/ μ l; urine protein: 1.5 g/d (57.5 mg/kg/d); serum albumin and creatinine were normal.	Prednisone for 6 mos; MMF for 1 yr. Response: both responded, urine protein reduced to 0.2–0.35 g/d.	Normal eGFR at 13 yrs
3	Female	Maternal grandfather and mother had hematuria and proteinuria; the mother's eGFR was normal at 45 yrs.	3	Urine RBC: 847/ μ l; urine protein: 0.95 g/d (63.3 mg/kg/d); serum albumin: 26.4 g/l; serum total cholesterol: 6.6 mmol/l; serum creatinine was normal.	Prednisone for 3 mos; CsA for 2 years; TAC for 6 months. Response: steroid resistant; responded to CsA and TAC (urine protein reduced to < 0.5 g/d) but failed to maintain long-term remission.	eGFR: 46 ml/min per 1.73 m ² at 13 yrs
4	Male	Mother: hematuria and proteinuria; younger brother: nephrotic syndrome (onset at age 5 yrs), eGFR normal at 6 yrs.	8	Urine RBC: 199/ μ l; urine protein: 2.98 g/d, serum total cholesterol: 8.14 mmol/l; serum albumin: 18.3 g/l; serum creatinine was normal.	Prednisone for 3 mos; TAC for 6 mos. Response: neither responded.	eGFR: 29 ml/min per 1.73 m ² at 14 yrs
5	Male	Maternal grandmother died of uremia; mother had hematuria and proteinuria; twin brother had a similar clinical course.	7	Urine RBC: 125/ μ l; urine protein: 3.87 g/d; serum albumin: 30.3 g/l; serum total cholesterol and creatinine were normal.	Prednisone for 6 mos; CTX administered twice (cumulative dose: 20 mg/kg). Response: neither responded.	eGFR: 58 ml/min per 1.73 m ² at 15 yrs

AS, Alport syndrome; CsA, cyclosporine A; CTX, cyclophosphamide; eGFR, estimated glomerular filtration rate; ESKD, end-stage kidney disease; IgAN, IgA nephropathy; MMF, mycophenolate mofetil; RBC, red blood cell; TAC, tacrolimus.

brother of case 5 carried the same mutation; his clinical presentation and disease course were similar to the proband's.

We performed double staining for IgA and KM55 in renal biopsy specimens from 5 patients with IgAN-AS. Cases 1 and 2 exhibited KM55 positive staining with an intensity of ++, case 3 was +, case 4 was $\pm$, and case 5 was negative (Figure 1, showing cases 1, 2, and 5).

Systematic Review

Search Results

The study selection process is illustrated in Figure 2. A total of 17 publications were included in the systematic review, comprising 12 full-text articles in English,^{4,5,12,17–25} 1 abstract in English,²⁶ and 4 full-text articles in Chinese.^{27–30} These studies were published between 2017 and 2025. Two of the publications reported on the same patient.

Clinical Features

Including our own series, a total of 81 patients from 72 families were identified. Detailed clinical, pathological, and genetic information of 76 patients with IgAN-AS extracted from 17 publications are provided in Supplementary Tables S1 and S2.

Detailed age data were available for 26 patients, with a median age at onset or diagnosis of 9.0 (7.0–20.3) years. Sex data were available for 81 patients, with 37 (45.7%) being male. All 64 patients with available data had microscopic hematuria, with 24 (37.5%) experiencing episodes of

macroscopic hematuria. Detailed individual proteinuria data were available for 25 patients, consisting of 19 children and 6 adults. Among these 25 patients, 15 (60.0%) exhibited nephrotic-range proteinuria, of whom 14 were children. Among these 81 patients, Yuan *et al.*⁵ reported an adult subgroup of 38 cases, in whom the median age at renal biopsy was 36.5 years (28.3–42.5), with a median 24-hour urinary protein excretion of 1.04 g (0.26–2.02) (individual-level data not provided, and proteinuria data were missing for 10 patients; among the remaining cases, 2 patients had nephrotic-range proteinuria). Overall, 73.7% of children (14/19) with IgAN-AS presented with nephrotic-range proteinuria, whereas 8.8% of adults (3/34) had nephrotic-range proteinuria ($P < 0.001$).

Kidney Biopsy Findings

A total of 92 renal biopsy results were obtained from 81 patients, including 7 patients who underwent a combined total of 18 biopsies (Supplementary Table S1). Only the results of the first renal biopsy were analyzed and are shown in Table 3.

Detailed immunofluorescence findings were reported for a subset of the included patients. IgA staining results were available for 58 patients, with intensities of + (8 cases, 13.8%), ++ (31 cases, 53.4%), +++ (18 cases, 31.0%), and ++++ (1 case, 1.7%). C3 staining results were available for 55

Table 2. Pathologic features and genetic findings of 5 children with IgAN-AS at our center

Patient	MEST-C classification	Immunofluorescence microscopy	Electron microscopy	Mutation	ACMG class and supporting criteria
1	MOEOS1TOC1	IgG (–), IgA (++), IgM (+), C3 (++) , C4 (–), C1q (–); $\alpha 3$ (IV) and $\alpha 5$ (IV) showed a normal staining pattern.	Segmental GBM thinning; chunky mesangial electron-dense deposits.	<i>COL4A5</i> c.4705C>T, p.R1569*, heterozygote	Likely pathogenic; PVS1,PM2
2	M1EOSOTOCO	IgG (–), IgA (++), C3 (+), C1q ($\pm$); segmental expression of $\alpha 5$ (IV) in the skin basement membrane.	Segmental thinning of the GBM; scanty, finely granular electron-dense deposits in the mesangial area.	<i>COL4A5</i> c.4821+2T>C, heterozygote	Pathogenic; PVS1,PS2,PM2
3	MOEOSOTOCO	IgG (–), IgA (++), IgM (++), C3 ($\pm$), C4 (+), C1q ($\pm$); weak and discontinuous expression of $\alpha 3$ (IV) and $\alpha 5$ (IV) in the GBM.	Irregular thickening and thinning of the GBM with splitting of the lamina densa; scattered mesangial and intramembranous electron-dense deposits.	<i>COL4A3</i> c.3725G>A, p.G1242D, heterozygote	Likely pathogenic; PM1,PM2,PP3,PP4
4	MOEOS1TOCO	IgG (–), IgA (++), IgM (++), C3 (+), C4 (–), C1q (–); absence of $\alpha 3$ (IV) and $\alpha 5$ (IV) expression in the GBM.	Diffuse irregular thickening and thinning of the GBM with lamellation and a basket-weave appearance of the lamina densa; scanty, chunky electron-dense deposits in the mesangial area.	<i>COL4A5</i> , deletion from exons 37 to 51, hemizygote	Pathogenic; PVS1,PM2,PP4
5	MOEOS1TOCO	IgG (–), IgA (++), IgM (+), C3 (–), C4 (–), C1q (–); absence of $\alpha 3$ (IV) and $\alpha 5$ (IV) expression in the GBM.	Irregular thickening and thinning of the GBM, with extensive lamellation of the lamina densa; chunky mesangial electron-dense deposits.	Not detected	Not applicable

ACMG, American College of Medical Genetics and Genomics; AS, Alport syndrome; GBM, glomerular basement membrane; IgAN, IgA nephropathy; MEST-C, Oxford classification scores; M, mesangial hypercellularity; E, endothelial hypercellularity; S, segmental sclerosis; T, tubular atrophy/interstitial fibrosis; C, cellular/fibrocellular crescents; PM, pathogenic moderate; PP, pathogenic supporting; PS, pathogenic strong; PVS, pathogenic very strong; $\alpha 3$ (IV), collagen type IV $\alpha 3$ chain; $\alpha 5$ (IV), collagen type IV $\alpha 5$ chain.

patients, with intensities of negative (17 cases, 30.9%), $\pm$ (2 cases, 3.6%), + (18 cases, 32.7%), ++ (14 cases, 25.5%), and +++ (4 cases, 7.3%).

MEST-C scores were available for 54 patients. M1, E1, S1, T1, T2, C1, and C2 lesions were observed in 14 (25.9%), 7 (13.0%), 22 (40.7%), 6 (11.1%), 1 (1.9%), 10 (18.5%), and 2 (3.7%) patients, respectively. For an additional 5 patients from 4 publications without formal MEST-C scoring, renal pathology was described qualitatively: 1 showed no significant mesangial hypercellularity, 1 mild, 2 moderate, and 1 unspecified degree of mesangial hypercellularity. None showed endocapillary hypercellularity, segmental sclerosis, or crescent formation (EOSOCO). One case showed 40% to 50% interstitial fibrosis with tubular atrophy (T1), and the remaining 4 had no or slight lesions (T0). After including these 5 cases, the total number of evaluable patients for E, S, T, and C lesions increased to 59. The frequencies of E1, S1, T1, T2, C1, and C2 lesions were 7 (11.9%), 22 (37.3%), 7 (11.9%), 1 (1.7%), 10 (16.9%), and 2 (3.4%), respectively (Table 3). M lesions remained evaluable in 54 patients only.

Electron microscopy results were available for 59 patients. Of these, 17 (28.8%) exhibited typical AS GBM features, including diffuse thickening or irregular thickness with lamellar splitting, tearing, and basket-weave patterns. Segmental or diffuse GBM thinning was the predominant finding in 41 (69.5%). One patient (1.7%) showed no apparent GBM abnormalities on the first biopsy but developed diffuse GBM thinning on repeat kidney biopsy. Patients with typical GBM changes of AS were associated with a

higher frequency of nephrotic-range proteinuria ($P = 0.04$), distinct C3 staining patterns ($P = 0.03$), and a predominance of X-linked inheritance in males ($P < 0.001$). No significant differences were observed in IgA staining intensity, gross hematuria, or Oxford MEST-C lesions (Table 4).

Intrafamilial Pathological Variability

In addition, multiple members with abnormal urinalysis underwent renal biopsy in 11 families. In 7 families, ≥ 2 members were diagnosed with IgAN-AS. Moreover, in 4 families, other affected members carrying the same P/LP *COL4A3/A4/A5* variant as their relatives with IgAN-AS exhibited different pathological types, including mesangial proliferative glomerulonephritis (IgG+, IgA–; $n = 1$), mesangial proliferative glomerulonephritis (IgM+, IgG–, IgA–; $n = 1$), and focal segmental glomerulosclerosis ($n = 2$).

Mutation

Among 81 patients with IgAN-AS, genetic testing results were available for 78, with 77 patients carrying P/LP *COL4A3/A4/A5* variants and 2 patients having a variant of uncertain significance (Supplementary Table S2). These 77 patients harbored a total of 66 distinct P/LP variants: 12 (18.2%) in *COL4A3*, 32 (48.5%) in *COL4A4*, and 22 (33.3%) in *COL4A5*. These variants comprised 44 missense mutations (37 involving glycine substitutions), 11 frameshift mutations due to deletions or insertions, 6 splicing variants, 2 nonsense mutations, 1 synonymous mutation, 1 large deletion, and 1 in-frame deletion (Table 3). Four

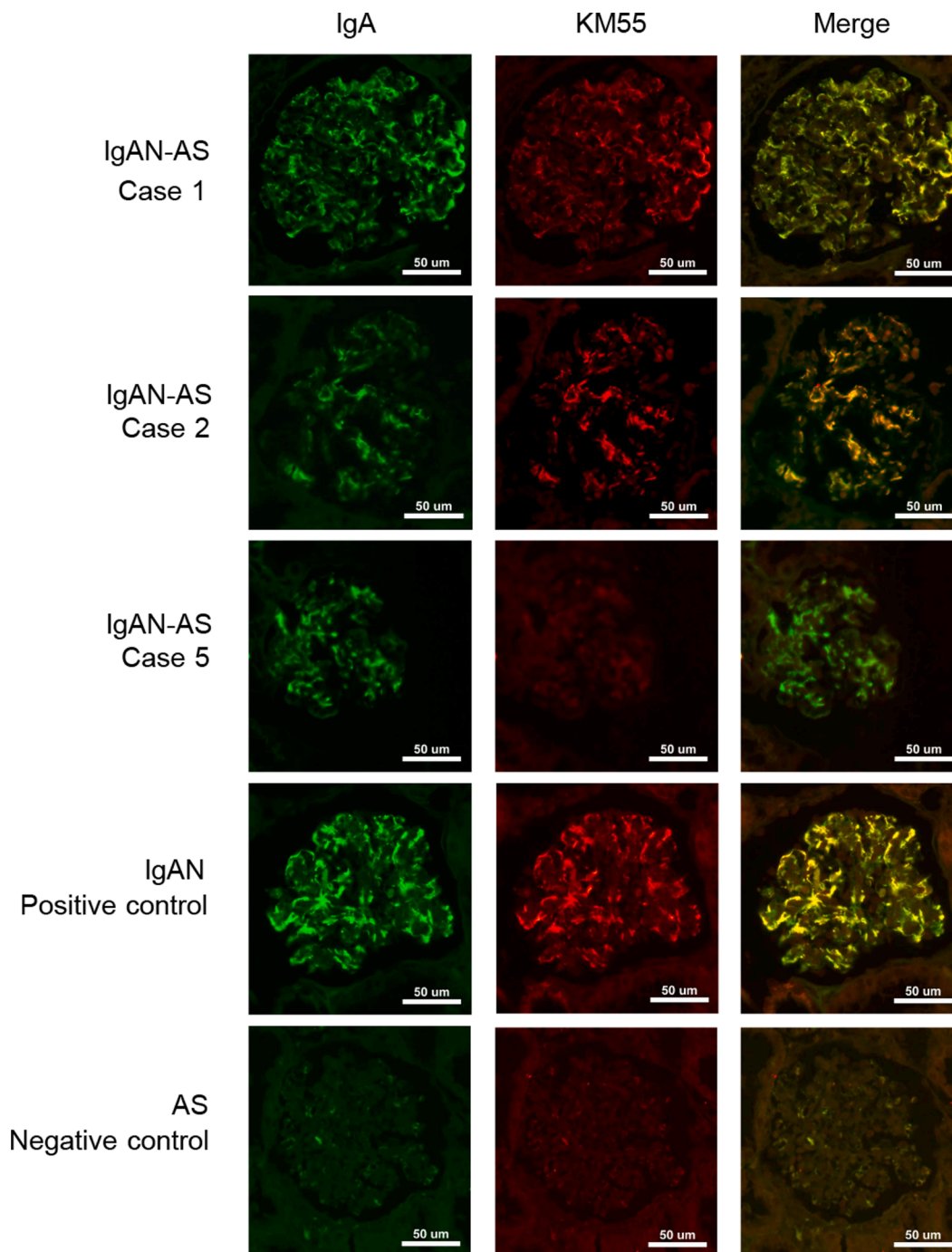


Figure 1. Glomerular deposition of galactose-deficient IgA1 in children with IgAN coexisting with AS (IgAN-AS). Double immunofluorescence staining with anti-IgA polyclonal antibody and anti-galactose-deficient IgA1 monoclonal antibody KM55 was performed on renal biopsy specimens from 5 children with IgAN-AS in our center. Images of 3 cases (cases 1, 2, and 5) are shown, corresponding to cases 1, 2, and 5 in [Tables 1](#) and [2](#). The positive control (primary IgAN) showed clear staining for both IgA and KM55, whereas the negative control (AS) was negative for both IgA and KM55 staining. (Bars = 50 μ m; original magnification, $\times 200$). AS, Alport syndrome; IgAN, IgA nephropathy.

variants (*COL4A3* c.3829G>A, *COL4A3* c.4510T>C, *COL4A3* c.4793T>G, and *COL4A4* c.4444delC) were identified in 2 different families each, whereas 1 variant (*COL4A4* c.3506_3528del) was found in 3 families.

Of the 77 patients with P/LP variants, 26 had IgAN-XLAS (13 males), 7 had IgAN-ARAS (3 homozygous cases), and 44 had IgAN with a single P/LP *COL4A3/A4*

variant (12 IgAN-COL4A3, 32 IgAN-COL4A4) ([Table 3](#)).

Immunological Features

Most studies reported only routine laboratory and pathological examinations, without further investigation of immunological features. Cambier *et al.* reported high plasma levels of specific immune complexes

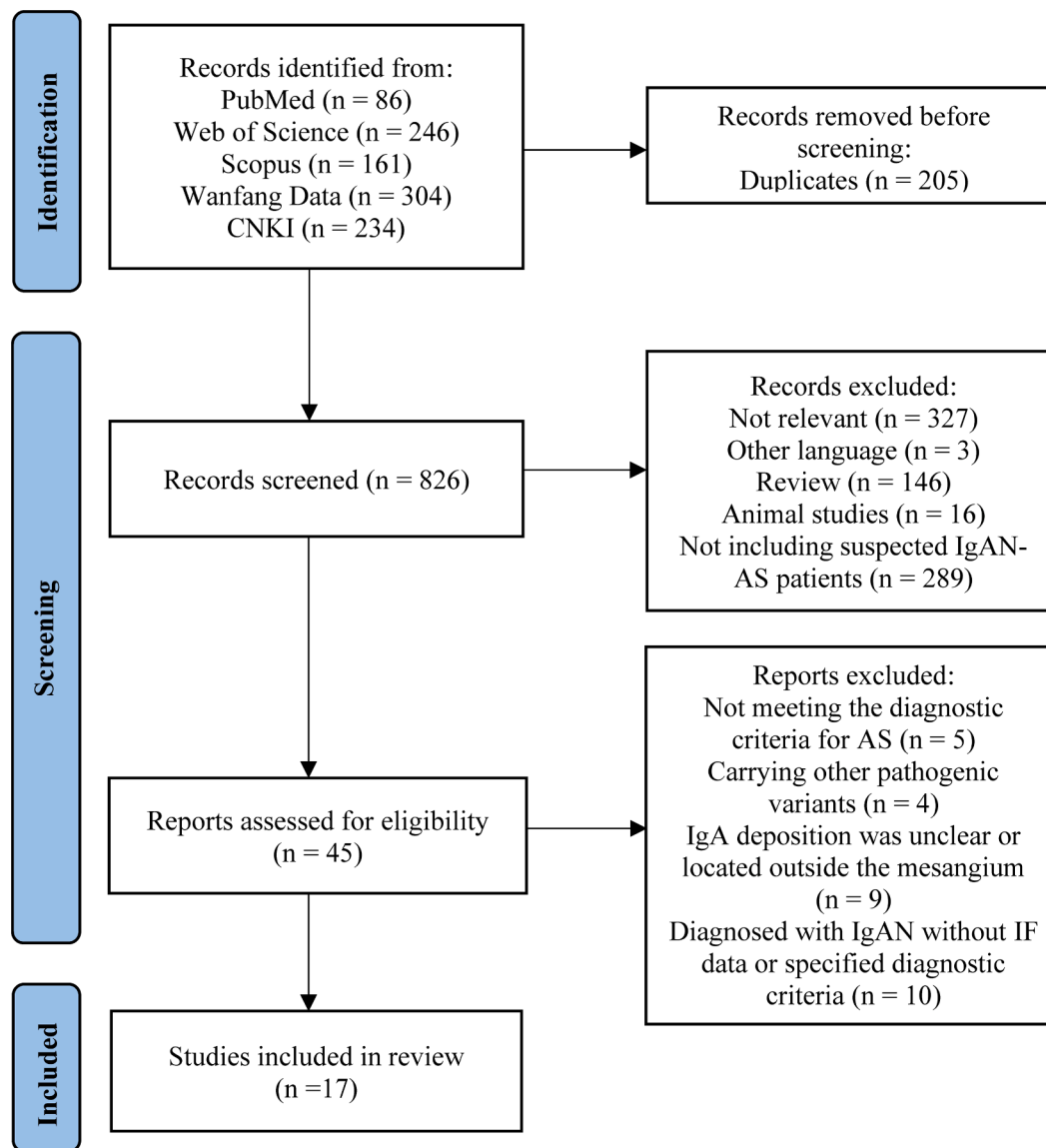


Figure 2. Flow diagram of the literature search and study selection process. AS, Alport syndrome; CNKI, China National Knowledge Infrastructure; IF, immunofluorescence; IgAN, IgA nephropathy.

(Gd-IgA1, IgG-IgA, soluble CD89-IgA, and free soluble CD89) in children with IgAN-COL4A3 ($n = 4$), comparable with those in children with IgAN-non-COL4A3.²² Conversely, Yuan *et al.* found that adult patients with IgAN-tGBM carrying P/LP COL4A3/A4/A5 variants ($n = 38$) had significantly lower plasma Gd-IgA1 levels than adult patients with IgAN without such variants, with levels similar to those of healthy controls.⁵

Treatment and Response

Treatment information was available for 25 patients. Two received neither immunosuppressive therapy nor RASB and both progressed to end-stage kidney disease at the ages of 32 and 34 years. Four received RASB alone, 1 had mild proteinuria at diagnosis (0.021 g/mmol) and a stable course, and 1 had preexisting CKD

G3. The remaining 2 had significant baseline proteinuria (800 and 960 mg/m²/d); the former showed minimal change in proteinuria and subsequently progressed to end-stage kidney disease at the age of 16 years, whereas the latter had proteinuria of 800 mg/m²/d at the 2-year follow-up, with estimated glomerular filtration rate declining from 92 to 70 ml/min per 1.73 m².

One patient received glucocorticoids alone, and 18 received glucocorticoids in combination with second-line immunosuppressants. Clinical, pathological, and prognostic features of the patients receiving immunosuppressive therapy are summarized in Table 5, with detailed treatment regimens and corresponding response data provided in Supplementary Table S3. Among the 19 patients who received immunosuppressive therapy, 13 had nephrotic-range proteinuria. Of the 14 patients evaluated for treatment response,

Table 3. Pathological and genetic characteristics of patients with IgAN-AS

Characteristics	n (%)
Pathological characteristics ^a	
Immunofluorescence	
IgA staining intensity (n = 58) ^b	+: 8 (13.8%), ++: 31 (53.4%), +++: 18 (31.0%), ++++: 1 (1.7%)
C3 staining intensity (n = 55) ^b	-: 17 (30.9%), ±: 2 (3.6%), +: 18 (32.7%), ++: 14 (25.5%), +++: 4 (7.3%)
Oxford classification	
M1 (n = 54) ^c	14 (25.9%)
E1 (n = 59) ^c	7 (11.9%)
S1 (n = 59) ^c	22 (37.3%)
T1/T2 (n = 59) ^c	7 (11.9%) / 1 (1.7%)
C1/C2 (n = 59) ^c	10 (16.9%) / 2 (3.4%)
Electron microscopy (n = 59) ^b	
Typical GBM changes of AS	17 (28.8%)
Segmental or diffuse GBM thinning	41 (69.5%)
Normal GBM thickness ^d	1 (1.7%)
Genetic characteristics	
Type of P/LP variants (n = 66) ^b	
Missense	44 (66.7%)
Frameshift	11 (16.7%)
Splicing	6 (9.1%)
Nonsense	2 (3.0%)
Synonymous	1 (1.5%)
Large deletion	1 (1.5%)
In-frame deletion	1 (1.5%)
Mode of inheritance (n = 77) ^b	
XLAS (male)	13 (16.9%)
XLAS (female)	13 (16.9%)
ARAS	7 (9.1%)
COL4A3 heterozygote	12 (15.6%)
COL4A4 heterozygote	32 (41.5%)

ARAS, autosomal recessive AS; AS, Alport syndrome; GBM, glomerular basement membrane; IgAN, IgA nephropathy; LP, likely pathogenic; MEST-C, Oxford classification scores: M, mesangial hypercellularity (M1: > 50% of glomeruli); E, endocapillary hypercellularity (E1: present); S, segmental glomerulosclerosis (S1: present); T, tubular atrophy/interstitial fibrosis (T1: 26-50%, T2: > 50%); C, cellular/fibrocellular crescents (C1: ≥ 1 glomerulus, C2: > 25% of glomeruli); P, pathogenic; XLAS, X-linked AS.

^aPathological characteristics were assessed based on the first kidney biopsy.

^bn indicates the number of patients for whom the corresponding variable was available.

^cn indicates the total number of patients evaluable for the corresponding available Oxford classification variable (M, E, S, T, or C), including both presence and absence categories (e.g., M0 + M1).

^dThis patient showed diffuse GBM thinning on a repeat kidney biopsy.

7.1% (1/14), 64.3% (9/14), and 28.6% (4/14) achieved CR, PR, and no remission, respectively. Majority of these patients were treated concomitantly with RASB (Supplementary Table S3). The presence of significant proliferative lesions (M1, E1, C1/C2) was significantly more frequent in patients who achieved CR or PR (90.0%, 9/10) than in those with no remission (25.0%, 1/4) ($P = 0.04$). Five patients were not evaluated because of low baseline proteinuria. Notably, remission could not be sustained under immunosuppressive therapy in case 3 from our center and in 2 patients reported in other studies.^{22,28}

Prognosis

Among 27 patients with follow-up data on kidney function and corresponding ages, the median age at

Table 4. Clinical, pathological, and genetic characteristics of patients with IgAN-AS stratified by the presence of typical GBM changes of AS

Characteristics	With typical GBM changes of AS	Without typical GBM changes of AS	P value
Male	10 (10/17, 58.8%)	14 (14/42, 33.3%)	0.07
Nephrotic-range proteinuria	8 (8/15, 53.3%)	6 (6/32, 18.8%)	0.04
Gross hematuria	8 (8/14, 57.1%)	13 (13/38, 34.2%)	0.13
Pathological characteristics			
IgA intensity			0.16
IgA+	1 (1/16, 6.3%)	6 (6/39, 15.4%)	
IgA++	7 (7/16, 43.7%)	24 (24/39, 61.5%)	
IgA+++	7 (7/16, 43.7%)	9 (9/39, 23.1%)	
IgA++++	1 (1/16, 6.3%)	0 (0/39, 0.0%)	
C3 intensity			0.03
C3-±	10 (10/15, 66.7%)	9 (9/39, 23.1%)	
C3+	2 (2/15, 13.3%)	15 (15/39, 38.5%)	
C3++	3 (3/15, 20.0%)	11 (11/39, 28.2%)	
C3+++	0 (0/15, 0.0%)	4 (4/39, 10.2%)	
Oxford classification			
M1	5 (5/12, 41.7%)	5 (5/38, 13.2%)	0.08
E1	0 (0/15, 0.0%)	4 (4/38, 10.5%)	0.57
S1	6 (6/15, 40.0%)	14 (14/38, 36.8%)	0.83
T1+T2	4 (4/15, 26.7%)	4 (4/38, 10.5%)	0.20
C1+C2	1 (1/15, 6.7%)	8 (8/38, 21.1%)	0.42
Mode of inheritance			
XLAS (male)	7 (7/14, 50.0%)	2 (2/40, 5.0%)	< 0.001
XLAS (female)	2 (2/14, 14.3%)	6 (6/40, 15.0%)	
ARAS	1 (1/14, 7.1%)	2 (2/40, 5.0%)	
COL4A3 heterozygote	2 (2/14, 14.3%)	5 (5/40, 12.5%)	
COL4A4 heterozygote	2 (2/14, 14.3%)	25 (25/40, 62.5%)	

ARAS, autosomal recessive AS; AS, Alport syndrome; GBM, glomerular basement membrane; IgAN, IgA nephropathy; MEST-C, Oxford classification scores: M, mesangial hypercellularity (M1: > 50% of glomeruli); E, endocapillary hypercellularity (E1: present); S, segmental glomerulosclerosis (S1: present); T, tubular atrophy/interstitial fibrosis (T1: 26-50%, T2: > 50%); C, cellular/fibrocellular crescents (C1: ≥ 1 glomerulus, C2: > 25% of glomeruli); XLAS, X-linked AS.

Percentages were calculated based on the number of patients with available data for each variable; therefore, denominators vary due to missing data.

last follow-up was 16.0 (10.0–24.0) years, and 11 (40.7%) reached the kidney end point. Kidney end point events were observed in 6 of 12 patients (50.0%) with ARAS or XLAS (male) and in 2 of 10 patients (20.0%) with XLAS (female) or heterozygous COL4A3/COL4A4 variants. With a mean follow-up time of 5.3 ± 3.4 years, kidney end point events were observed in 6 of 14 patients (42.9%) with nephrotic-range proteinuria at onset or diagnosis and in 2 of 9 patients (22.2%) without nephrotic-range proteinuria.

DISCUSSION

This study provides the first comprehensive systematic review of IgAN-AS, summarizing data from a total of 81 patients, including 5 cases from our institutional pediatric series. Microscopic hematuria was universally observed, with nephrotic-range proteinuria more common in children than adults. The MEST-C scores

Table 5. Characteristics of patients with IgAN-AS treated with immunosuppressive therapy

Patient	Study	Renal pathology results	Age at treatment, yrs	Proteinuria	Treatment response	Age of prognosis, yrs	Prognosis
1	Our center, case 1	MOE0SOTOC1	15	0.61 g/d	NA	24	CKD G1
2	Cui <i>et al.</i> ²⁹	M1E0SOTOCO	7	> 50 mg/kg/d ^a	PR	8	CKD G1
3	Cui <i>et al.</i> ²⁹	M1E0SOTOCO	6	125 mg/kg/d ^a	PR	7	CKD G1
4	Vischini <i>et al.</i> ²³	M1E1SOTOC1	9	uPCR: 4.65 g/g ^a	PR	16	ESKD
5	Li <i>et al.</i> ²⁴	MOE0S1T1CO	9	4.4 g/d ^a	NR	9	CKD G1
6	Liu <i>et al.</i> ²⁷	Mild diffuse mesangial proliferation, EO, SO, TO, CO.	22	0.79 g/d	NA	36	CKD G3b
7	Liu <i>et al.</i> ²⁷	Mild to moderate, diffuse mesangial proliferation, EO, SO, TO, CO.	26	0.63 g/d	NA	34	CKD G1
8	Cambier <i>et al.</i> ²²	M1E1SOTOC1	8	uPCR: 0.4 g/mmol ^a	PR	17	CKD G3a
9	Cambier <i>et al.</i> ²²	M1E1S1T0C2	14	uPCR: 1.57 g/mmol ^a	PR	21	ESKD
10	Cambier <i>et al.</i> ²²	M1E1S1T0C2	10	uPCR: 0.65 g/mmol ^a	PR	14	CKD G2
11	Liu <i>et al.</i> ²⁸	Moderate diffuse mesangial cell proliferation, EO, SO, TO, CO.	7	6.27 g/d ^a	PR	9	CKD G1
12	Aksenova <i>et al.</i> ²⁶	M1E0S1T0CO	13	1.45 g/m ² /d ^a	NR	17	CKD G1
13	Xu <i>et al.</i> ²⁵	M1E0S1T1CO	23	2.63 g/d	CR	24	CKD G1
14	Yuan <i>et al.</i> ³⁰	M1E0SOTOCO	5	0.337 g/d	NA	8	CKD G1
15	Shao <i>et al.</i> ¹²	MOE0S1T1C1	2	uACR: 0.15 g/g	NA	6	CKD G1
16	Our center, case 2	M1E0SOTOCO	8	57.5 mg/kg/d ^a	PR	13	CKD G1
17	Our center, case 3	MOE0SOTOCO	3	63.3 mg/kg/d ^a	PR	13	CKD G3a
18	Our center, case 4	MOE0S1T0CO	8	2.98 g/d ^a	NR	14	CKD G4
19	Our center, case 5	MOE0S1T0CO	7	3.87 g/d ^a	NR	15	CKD G3a

AS, Alport syndrome; CKD, chronic kidney disease; CR, complete remission; ESKD, end-stage kidney disease; IgAN, IgA nephropathy; MEST-C, Oxford classification scores: M, mesangial hypercellularity score; E, endocapillary hypercellularity score; S, segmental glomerulosclerosis score; T, tubular atrophy/interstitial fibrosis score; C, crescent score; NA, not applicable; NR, no remission; PR, partial remission; uACR, urine albumin-to-creatinine ratio; uPCR, urine protein-to-creatinine ratio.

^aNephrotic-range proteinuria.

Patient 1 was treated with glucocorticoids alone, whereas patients 2–19 received glucocorticoids and other second-line immunosuppressants.

showed M1 lesions in 25.9%, E1 in 11.9%, S1 in 37.3%, T1+T2 in 13.6%, and C1+C2 in 20.3%. Electron microscopy revealed GBM abnormalities, including thinning in 69.5% and AS-specific changes in 28.8%. Genetic analysis identified 66 distinct *COL4A3/A4/A5* pathogenic variants. Inconsistent Gd-IgA1 detection results suggest complex pathogenesis. During immunosuppressive treatment, majority of patients achieved CR or PR. Notably, 40.7% of the patients reached the end point of kidney outcome with a median age at last visit of 16.0 years. These findings contribute to a more systematic understanding of IgAN-AS, suggest considerable heterogeneity, and support the need for multidimensional evaluation to optimize diagnosis and stratification.

The researchers hypothesized that the association between AS and immune-mediated glomerular diseases suggests that GBM defects caused by pathogenic *COL4A3/A4/A5* variants may predispose individuals to immune-mediated glomerular injury.³¹ Although reports of IgAN-AS have increased, the frequency of coexistence remains unclear. Predicted pathogenic *COL4A5* variants affect approximately 1 in 2320 individuals and heterozygous *COL4A3/A4* variants about 1 in 106,³² implying that approximately 1% of patients with IgAN may incidentally coexist with AS. In sporadic IgAN, the prevalence of AS ranges from 0.49% to 11.1%,^{5,22} whereas in familial IgAN it is 66.7% to

75.0%^{4,27}; the lower detection in the 2 sporadic cohorts is likely because of limited genetic screening (Yuan *et al.*⁵ reported 0.49% and our center observed 0.90%). This study analyzed families in which multiple members underwent kidney biopsy and found that within the same family, >1 individual carrying the same variant was diagnosed with IgAN. In conclusion, IgAN-AS may not merely be a coincidental coexistence of 2 diseases.

However, some family members carrying the same variant showed other pathological types (such as focal segmental glomerulosclerosis or IgA-negative glomerulonephritis). Furthermore, our data revealed no significant association between the severity of GBM structural damage and the intensity of mesangial IgA deposition. In addition, it is generally believed that AS rarely involves immune complex deposition.³³ These findings argue against a purely passive mechanical entrapment mechanism secondary to GBM defects. Instead, whereas GBM injury may create a permissive microenvironment, the initiation and progression of IgAN-AS likely result from a complex interplay between structural vulnerability and independent immunopathogenic mechanisms.

This complex interplay likely underlies the marked clinicopathological heterogeneity observed in IgAN-AS. Most patients showed marked IgA deposition ($\geq 2+$ in 86.2 %) and frequent C3 codeposition (65.5%),

distinguishing IgAN-AS from pure AS, where immune complexes are rare. Low rates of endocapillary hypercellularity (E1, 11.9%) and crescents (C1/C2, 20.3%) were noted, unlike the higher E/C scores of reported primary IgAN cohorts.^{34,35} Ultrastructurally, typical AS features (e.g., lamellation) were less common (28.8%) than diffuse or segmental thin GBM, which predominated (69.5%). Overall, the intensities of IgA/C3 deposition, the distribution of MEST-C lesions, and the spectrum of GBM abnormalities collectively reflect the pathological heterogeneity of IgAN-AS.

In addition to pathological heterogeneity, IgAN-AS exhibits considerable genetic diversity. Among 77 patients with P/LP *COL4A3/A4/A5* variants, a total of 66 distinct variants were identified, mainly including missense, frameshift, and splicing mutations, which were distributed across the 3 genes. Some variants recurred in multiple families; however, most were unique, reflecting a wide spectrum of genetic alterations underlying the disease. This extensive mutational heterogeneity may contribute to variability in GBM structural vulnerability and modulate the susceptibility to immune-mediated injury.

Immunological features in IgAN-AS varied across studies and age groups. In our single-center series, 3 of 5 pediatric patients with IgAN-AS showed positive Gd-IgA1 staining. Cambier *et al.*²² reported elevated plasma levels of Gd-IgA1 and related immune complexes in children with IgAN-COL4A3, whereas Yuan *et al.*⁵ observed significantly lower plasma Gd-IgA1 levels in adult patients with IgAN-AS compared with IgAN without *COL4A3/A4/A5* variants. Because neither study established a clear cutoff for positivity, the true prevalence of elevated circulating Gd-IgA1 remains difficult to determine. Nonetheless, these findings suggest that Gd-IgA1 levels may not be consistently elevated in IgAN-AS, particularly in adults. Importantly, this does not imply that IgA deposition is purely secondary; as previously discussed, it is unlikely to be explained solely by passive mechanical entrapment due to GBM defects. Furthermore, >20% of patients with primary IgAN do not exhibit elevated Gd-IgA1,³⁶ and N-glycosylation abnormalities likely contribute to pathogenesis.³⁷ Current Gd-IgA1-based assays may not capture all disease mechanisms. These observations underscore the need for more sensitive and specific biomarkers or detection methods for IgAN to refine disease stratification in IgAN-AS.

Treatment response during immunosuppressive therapy in IgAN-AS appeared heterogeneous but generally favorable among patients with active immune lesions, suggesting that immune activity remains an important therapeutic target even in the presence of

structural GBM abnormalities. However, it should be noted that most patients were treated concomitantly with RASB. Although we limited the analysis to patients with higher baseline proteinuria (≥ 1.0 g/d or ≥ 1.0 g/g) to enhance the clinical relevance of the treatment response assessment, the findings should be interpreted with caution because of the potential confounding effect of RASB therapy. In addition, sustained remission was not universal; several patients experienced relapse despite ongoing therapy, indicating that GBM fragility or irreversible structural damage may limit the long-term durability of treatment response.

In this systematic review, 40.7% of 27 patients with IgAN-AS reached the kidney end point at a median age of 16 years at last follow-up. In comparison, an international adult IgAN cohort (used to develop the International Risk-Prediction Tool) reported a 5-year incidence of 50% estimated glomerular filtration rate decline of 15.1%, and pediatric IgAN is generally associated with more favorable outcomes than adult disease.^{34,35} In AS, male XLAS typically exhibits poorer prognosis, with a median kidney survival age of 25 to 30 years, whereas autosomal dominant AS and female XLAS often have median kidney survival beyond 55 years.^{38–42} Although differences in study design, end point definitions, and follow-up duration limit direct comparison, these observations may suggest a relatively less favorable renal outcome in IgAN-AS.

This study has several limitations. First, because of the retrospective design and heterogeneity of the included studies, several issues prevented a comprehensive analysis and potentially introduced bias, including the following: incomplete data, variability in biopsy protocols, the unavailability of immunological indicators (such as Gd-IgA1 levels) for all cases, and nonstandardized treatment decisions. Second, despite assembling the largest IgAN-AS cohort to date, small sample sizes limited the statistical power for subgroup analyses and precluded definitive conclusions on genotype-phenotype correlations and survival analysis. Third, potential geographical bias from predominantly Chinese data may limit generalizability. Fourth, the diagnostic differentiation between primary IgA and incidental mesangial IgA deposition in patients with AS remains a methodological challenge. This is because of several factors including the following: (i) there are currently no widely accepted biomarkers with high sensitivity and specificity for IgAN; and (ii) majority of patients have not undergone additional immunological profiling, and (iii) the absence of clearly defined positivity thresholds complicates interpretation. Consequently, we relied on the presence of dominant or codominant mesangial IgA deposition as the diagnostic criterion for IgAN in this study.

In summary, this systematic review establishes IgAN-AS as a distinct and heterogeneous “dual-pathology” entity, defined by a complex interplay of immune-mediated injury (IgAN) and structural GBM abnormalities (AS). Our findings suggest a relatively poor renal prognosis. Critically, these results indicate the need for a tailored therapeutic approach: immunosuppressive therapy might benefit patients with active immune lesions, whereas those dominated by chronic structural damage may primarily require optimal supportive therapy. IgA deposition in IgAN-AS is unlikely to be explained solely by passive mechanical entrapment due to GBM defects; rather, its initiation and progression likely involve independent immunopathogenic mechanisms, and biomarkers with high sensitivity and specificity for IgAN will be essential for disease stratification. Larger prospective studies are needed to validate these findings and guide precision management.

DISCLOSURE

All the authors declared no competing interests.

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DATA AVAILABILITY STATEMENT

Data that support the findings are included within the manuscript.

SUPPLEMENTARY MATERIAL

[Supplementary File \(PDF\)](#)

Table S1. Clinicopathological features of 76 patients with IgAN-AS identified in the literature.

Table S2. Genetic features of 76 patients with IgAN-AS identified in the literature.

Table S3. Detailed treatment information of the 19 patients with IgAN-AS who received immunosuppressive therapy.

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